STANCE: Environment Implementation Report

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1 Overview

This submission contains the **simulation environment**: a planar MuJoCo leg with a custom deformable-ground model, wrapped as a Gymnasium environment and registered as `StanceAnkle-v0` (training ground) and `StanceAnkleTest-v0` (held-out ground). **To run the demo please see** `README.md`

2 Code explanation

2.1 Organisation

One owner per file. These three files meet at three fixed interfaces:
(**ground** $\rightarrow$ **environment**, **leg XML** $\rightarrow$ **environment** and **observation/action ordering**).

File	Responsibility	Author
`model/leg.xml`	MuJoCo model: 3 passive foot DOFs, actuated `ankle_hinge` ($\pm 20^\circ$, ± 150 N m), passive `pylon_slide` (spring + damper), 70 kg body mass; native collisions off.	Muthaiah
`stance_env/ground.py`	Force law, per-episode material sampling, persistent dent.	Mukul
`stance_env/ankle_gym_env.py`	`AnkleEnv`: reset, step, observation, reward, termination, rendering.	Vishal
`demo.py`	Demo file that can run visual renders of randomized episodes	All 3

2.2 Leg model (`leg.xml`)

MuJoCo was used to develop a planar prosthetic leg with five linked bodies: `foot` $\rightarrow$ `shank` $\rightarrow$ `pylon` $\rightarrow$ `upper_leg` $\rightarrow$ `body_mass`. Three passive foot joints (`foot_x`, `foot_z`, `foot_pitch`), an actuated ankle hinge ($\pm 20^\circ$, ± 150 Nm), and a passive spring-damper pylon slider are among the 5-DOFs defined by the leg.xml file.

Custom ground-force computations and applications are made possible by heel and toe reference locations. The standing height of the model is 0.90 m and its total mass is 72.41 kg.

Component	Mass (kg)	Purpose
Human body (`body_mass`)	70.00	Represents the user's body loading
Foot (`foot_geom`)	0.80	Prosthetic foot
Shank (`shank_geom`)	1.20	Lower prosthetic structure
Pylon (`pylon_geom`)	0.40	Connecting shaft with passive compliance
Upper link (`upper_leg_geom`)	0.01	Structural link from pylon to body mass
Total	**72.41**	Combined simulation mass

2.3 Ground model (`ground.py`)

MuJoCo's built-in contact solver has been turned off. Instead, the heel and toe each are acting as a spring and damper that push back once the foot sinks below the surface. At depth d and sinking speed $\dot{d}$, the upward force F_n and friction F_t are:

$$F_n = \max\big(k(d)\,d + c\,\dot{d},\ 0\big), \qquad k(d) = k_0\big(1 + \alpha\, d/d_{\text{ref}}\big), \qquad F_t = -\mu F_n \tanh(v_t/v_\epsilon).$$

Here k_0 is the base stiffness, α how fast it stiffens with depth ($d_{\text{ref}} = 2$ cm), c the damping, μ the friction coefficient and v_t the sliding speed; $\max(\cdot, 0)$ prevents an "attractive" reaction force from the ground, and tanh smooths friction near $v_t = 0$ ($v_\epsilon = 1$ mm/s). Here are a few salient features of our ground model:

- **Permanent dents:** The ground gives way past a yield force of f_y. The normal force remains f_y and a persistent dent appears in the ground.
- **A new ground every episode:** Each new episode, a new set of k_0, α, f_y, μ and a damping ratio ζ are drawn for each contact point. Then damping is calculated as: $c = 2\zeta\sqrt{k_0 m}$ (here m: leg mass). Note: On select surfaces, we have applied a damping cap to keep the leg from bouncing.
- **Unseen test grounds:** test values of k_0, α, f_y and μ lie outside the training ranges.

Changes: damping c was first drawn on its own, so hard grounds could be almost undamped and very bouncy while soft ones were damped far more. Now, we are drawing the ratio ζ and calculating c from it.

Limitations: 1.The foot touches the ground at only two independent points, unlike real soil. 2.The damping cap leaves some grounds less damped than they are drawn.

2.4 Environment Logic (`ankle_gym_env.py`)

`AnkleEnv` exposes the MuJoCo leg (`leg.xml`) and our ground model (`ground.py`) as a Gymnasium environment. Each `step()` rescales the agent's three outputs to a target angle, stiffness and damping, then runs ten $0.5\,\text{ms}$ physics substeps (a

$200\,\mathrm{Hz}$ controller). Each substep refreshes the heel and toe states, applies their ground forces, sets the impedance torque $\tau = K(\theta_d - \theta) - B\dot{\theta}$ (limited to $\pm 150\,\mathrm{N\,m}$) and advances MuJoCo. The step then logs peak forces, appends the sensor reading to a 10-step history and returns the observation, reward and flags.

Element	Implementation
Action	$a \in [-1,1]^3 \mapsto (\theta_d, K, B)$; $K \in [5, 500]$ N m/rad, $B \in [0.5, 10]$ N m s/rad, $\theta_d \in [-20^\circ, 20^\circ]$.
Observation	Last 10 sensor vectors, flattened to 110 values. Each vector has 11 channels: ankle angle, ankle rate, motor torque, vertical GRF, pylon compression, shank tilt, shank tilt rate, vertical acceleration, and the previous action (3 values). Ground parameters are never exposed.
Reward	$r = 1 - 2.94 \cdot \max\left(0, \frac{F_{\text{toe}}}{mg} - 1.58\right)^2 - 0.5\|a - a_{\text{prev}}\|^2 - 25 \cdot [\text{fall}]$ where F_{toe} is the peak toe force over the 10 substeps.
Termination	*Terminated* if shank tilt $< -15^\circ$ (backward) or $> 35^\circ$ (forward), or body height $<$ 70% of standing height (collapse); *truncated* at 100 steps.
Reset	New ground; heel exactly on the surface; forward speed 0.8–1.2 m/s, downward speed 0.10–0.50 m/s, leg tilt -8° to 4°, ankle angle $\pm 5^\circ$.
Info	Peak GRF (body weights), tilt, height, ground parameters and dent depth, so fall rate by failure type can be logged.

3 Key Libraries Used in this Submission

Library	Used for	What we did to make it work
MuJoCo (`mujoco`)	Rigid-body dynamics, `mj_step`, `mj_applyFT`, rendering.	Native collisions disabled (`contype`, `conaffinity` set to 0) and replaced by our own contact law, since MuJoCo's contact model cannot represent depth-stiffening, permanent dents or yield.
Gymnasium	`Env` API, `Box` spaces, registration.	Two environment ids registered so training and held-out ground cannot be mixed up.

Other libraries:NumPy for sampling, vector maths.

4 Reflections and lessons learned

4.1 Changes from the proposal

Item	Proposal	Implementation
Action scaling	Ranges for θ_d, K, B (no scaling specified)	K and B mapped on a *log* scale (a linear map leaves soft stiffnesses nearly unreachable); θ_d linear.
Leg size	70 kg mass, 0.85 m leg	Built model: 72.4 kg total, 0.90 m standing height (foot, shank and pylon add mass and height).
Start pose	Random ankle angle and leg tilt	Heel placed on the surface. Center of mass is then rerolled such that it lies between the heel and toe.
Forward speed	0.8–1.4 m/s	0.8–1.2 m/s. This is since if the movement is too fast, the leg collapses.
Tilt limit	$\pm 15^\circ$ from vertical	15° backward, 35° forward, since the shank naturally tilts forward during stance.
Impact penalty	Total ground force F/mg	Peak *toe* force, as total force barely responds to K; $\lambda = 1.58$ BW, $w_1 = 2.94$.
Fall penalty	None (episode just ends)	-25 once on a fall; otherwise falling early beat paying the impact penalty.

4.2 Lessons learned

(1) **Terminated vs. truncated:** time-limit endings must be *truncated*, or the critic learns that surviving is worthless.
(2) **Applied forces persist:** `qfrc_applied` is not automatically cleared by MuJoCo and must be zeroed every step.
(3) **Leg tilt:** needed to read data from the **shank's** world rotation, not from the foot's `foot_pitch`.
(4) **Bouncing was numerical:** contact forces relies on position but we were updating it with a one-step lag. The ground damping was also too strong for the light foot, so the simulation injected too much energy into the system. This led to the foot bouncing around across the surface. Halving the timestep to $0.5\,\mathrm{ms}$ and capping the damping, cut the bouncing from 14% of episodes to near 0 over N episodes under a [random/fixed] policy%.
(5) **Unwinnable starts:** poses with the center of mass behind the heel almost always fell backward. We now reject those poses and limit forward speed to 0.8–1.2 m/s.
(6) **Wrong impact measure:** total ground force barely changed with ankle stiffness. Toe force responds strongly to our changes in actions, so we measure that instead and plan to penalize large impacts on the toe.
(7) **Falling was too cheap:** a fall only cost about 20 steps of survival reward, so the agent learned to fall early to escape the impact penalty. Changed to a tentative -25 fall penalty (might change after further testing).